NIVESHA AI STUDIO SPECIFICATION

DOC-ID: NV-DES-2026-v1

CLASSIFICATION: INTERNAL

Personal Finance Coach & Living Money Model

Comprehensive Technical Architecture & Component Specifications

1. System Architecture Overview

Nivesha operates as an agentic personal wealth management ecosystem. Rather than treating transactions as static log lines, the system constructs a **Living Financial Knowledge Graph** that synchronizes real-time ingestion, deterministic cash flow forecasting, Monte Carlo scenario simulation, and strict regulatory compliance guards.

2. Multi-Modal Ingestion & Normalization Pipeline

The ingestion layer processes raw statements from disparate formats (bank PDFs, SMS payloads, UPI confirmation receipts, CSV extracts) into a normalized schema using deterministic rules backed by small-footprint extraction models.

```
{
  "transaction_id": "tx_98234123",
  "raw_text": "INR 499.00 debited from A/C XX1042 on 12-Oct-26 to Netflix India via UPI Ref 429
  "amount": 499.00, "currency": "INR", "type": "DEBIT",
  "merchant": "Netflix India", "category_primary": "Entertainment",
  "category_secondary": "Streaming Services", "is_recurring": true
}
```

3. Adaptive Correction Memory Engine

When a user overrides an automatic categorization (e.g., reclassifying `Amazon Retail` from `Shopping` to `Office Supplies`), the correction loop creates an active learning tuple `(merchant_raw, amount_range, time_pattern, old_cat, new_cat)`. Explicit user rules take absolute precedence over global defaults.

4. Living Financial Graph & Recurring Entity Detection

The graph tracks interlinked nodes: `[User] → [Income Streams] → [Accounts] → [Liabilities] → [Recurring Subscriptions] → [Goals]`. Subscription intervals are detected by clustering debit frequencies ($T \in [28, 32]\text{ days}$). Price-creep alerts trigger when $A_{\text{current}} > A_{\text{baseline}} \times 1.03$.

5. Website Design & UI/UX System Architecture

The frontend web interface is built using a modern, dark-slate glassmorphism visual design to convey premium security and intelligent automation.

- **Design System Palette:** Deep Slate Background (`#020617`), Vibrant Emerald Accents (`#10b981`), Warm Amber Highlights (`#f59e0b`), Slate Borders (`#1e293b`).
- **Layout System:** Sticky glassmorphism header navigation, full-width responsive container, interactive 3-column dashboard grid layout with smooth transitions.
- **Typography Stack:** Clean Sans-Serif font stack with geometric monospace numerical styling (`font-mono`) for real-time balances, calculations, and timeline metrics.
- **Micro-Interactions:** Real-time slider feedback, state transition animations on scenario toggles, and pulse states for active OCR/SMS statement parsing.

6. Stochastic Scenario & What-If Simulation

The financial simulator runs 1,000 Monte Carlo stochastic runs combining discretionary spending volatility (σ_S), inflation (π), and liquid asset yields (r) to compute probability of goal completion:

$$S_{t+1} = S_t \cdot (1 + r_{monthly}) + M \cdot N(1, \sigma_S) - Shock_t$$

7. UI/UX Wireframe & Component Layout Matrix

UI Component	Visual Treatment & Pattern	Interactive UX Behavior
Header & Nav	Backdrop blur (`slate-950/80`), gradient logo badge.	Quick jump to Simulator, Parser Sandbox, and Guardrails sections.
Simulator Sliders	Dual-accent slider controls with live numerical pill counters.	Re-computes monthly net surplus and goal completion date on drag.
Nivesha Brain Card	Dynamic glass panel with accent borders (Emerald/Amber).	Displays real-time velocity warnings and tailored persona recommendations.
Parser Sandbox	Dark terminal block with tabbed sample inputs.	Shows extraction state animation and auto-learned rule output box.

8. Regulatory Guardrails & Compliance Filter

An automated non-advisory filter middleware inspects all LLM outbound drafts against regex and semantic rules. Any explicit security recommendation (e.g., *"Buy HDFC stock"*) is intercepted and reframed into educational scenario guidance.

9. Core API Specification

Endpoint	Method	Payload & Description
`/api/v1/ingestion/parse`	**POST**	Accepts raw SMS/PDF text strings; returns normalized merchant entity and category classification.
`/api/v1/simulation/run`	**POST**	Accepts baseline profile and scenario deltas (rent hikes, salary changes); outputs 24m trajectory.
`/api/v1/agent/sweep`	**POST**	Triggers automated liquidity sweeps when checking balance exceeds fixed monthly expenses by 1.5x.

10. Database Schema (Core Tables)

The persistent layer uses relational PostgreSQL tables for users, financial profiles, immutable transaction ledgers, user correction memory rules, and priority goals.

Engineering Implementation Status:

All specifications and UI/UX design components are fully validated for AI Studio deployment. Zero external dependencies or uncompiled math blocks. Ready for production integration.